

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 128

Does R2 show up as an OSPF neighbor on R1? No

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

R2 is not forming OSPF adjacency on the passive interface, so traffic uses the alternate path through R1 and R3.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

74. serial link cost 64 and FastEthernet cost 10.

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The modern interfaces are faster than 100 Mbps, changing reference bandwidth allows OSPF to calculate costs more accurately.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The sum of all outgoing interface costs along the path. cost = $781 + 64 = 845$.

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

The new accumulated cost is 156, because all serial interfaces now use bandwidth 128 kbps, it will increase the interface cost.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The route to 192.168.3.0/24 now goes through R2 because the OSPF cost on R1's direct link to R3 was manually increased. This made the total cost of the direct path higher than the path through R2, so OSPF chose the lower-cost route through R2.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

_ The router ID is important because it gives each OSPF router a unique identity in the network. It helps routers form correct neighbor relationships and keeps OSPF stable during communication.

2. Why is the DR/BDR election process not a concern in this lab?

_ Because the routers are connected by serial point-to-point links. In point-to-point networks, OSPF does not need DR or BDR election.

3. Why would you want to set an OSPF interface to passive?

_ A passive interface stops OSPF hello packets from being sent to devices like PCs. This reduces unnecessary traffic but the network can still be advertised to other routers.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

_ OSPF becomes hierarchical because it uses different areas with Area 0 as the backbone. This helps reduce routing updates and makes large networks easier to manage.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

_ network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

_ The ~~OSPF~~ route is chosen because OSPF has lower administrative distance than RIP and EIGRP external, so the router prefers OSPF first.